COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	7/15/26	**DATE SUBMITTED:**	7/15/26
NAME:	Prince Benjamin S. Villaruel	**INSTRUCTOR:**	Engr. Jimlord Quejado

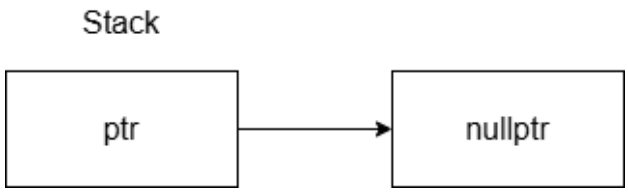
ACTIVITY NO. 2.1 Memory Diagram Exercise

1. PROCEDURE OUTPUT

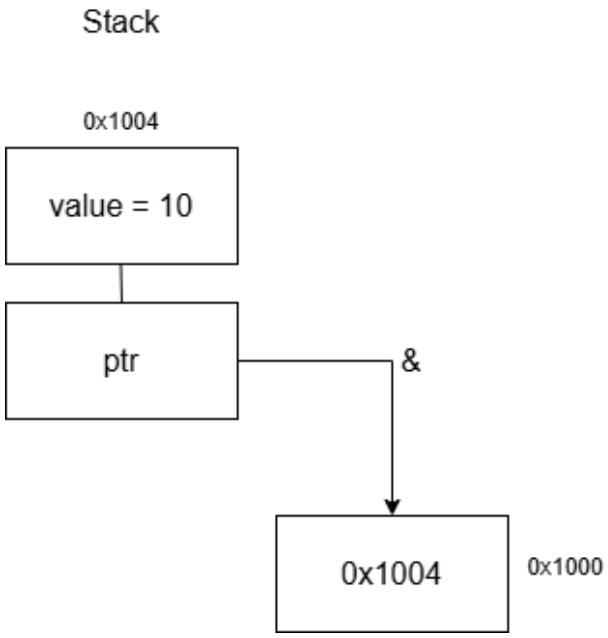
What procedures did you perform?

Draw the memory diagram for the following:

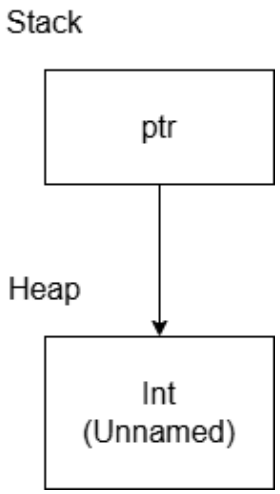
1. **int *ptr = nullptr;**



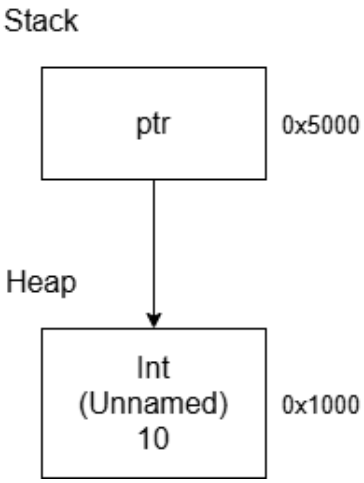
2. **2. int value = 10;**
   int *ptr= &value;



3. **int *ptr = new int;**



4. **int *ptr = new int(10);**



5. **Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.**

In simple terms, both the constructor and destructor mark an object's start and end, wherein constructor automatically runs when an object is created and destructor releases whatever the constructor acquired when the object's deleted, preventing any leaks. The copy constructor and copy assignment operator both deal with duplicating an object, but they step in at different moments. A copy constructor runs when a new object is made from an existing one, nothing old to clean up yet. An assignment operator runs when an existing object is given new values from another object, so it has to free its old memory first before copying in the new.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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